# QCert: Verifying and Extending Quantum Circuit Completeness in Agda and Lean

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A set of circuit equations is complete if any two circuits denoting the same linear map can be derived from each other using these equations, and minimal if no equation can be derived from the others. Establishing these properties takes long case analyses spanning hundreds of pages, making the details hard to check by hand. To this end, we extend the previous formalization framework to QCert, now available in both Agda and Lean, and use it to formalize the main circuit completeness theorems to date. This includes sixteen results, fifteen on completeness and one on minimality, over finite gate sets such as Clifford, CNOT-dihedral, Clifford+T, Clifford+CS, real Clifford+CH, and Toffoli–Hadamard, on qubits and on qudits of every odd prime dimension. QCert divides the work between humans and LLM coding agents. We first built its framework by specifying the syntax, semantics, and group presentations. Then, each paper's main theorem can be expressed uniformly in this framework. Next, an agent took over and formalized the proofs in Agda, following the papers' arguments. Agda checked every proof, so a human need only check the relation table, the gate semantics, and the theorem statement for each paper. Overall, QCert confirms the main results of each work and identifies gaps in two of the existing proofs. QCert is also useful for solving new problems. Using it, we generalize the completeness of Toffoli–Hadamard circuits from three qubits to any number of qubits. Its ability to do verification at scale makes QCert well suited to verifying quantum circuit completeness in higher dimensions.

## 1 Introduction

Rewriting of quantum circuits is a ubiquitous tool in circuit optimization [28, 35, 36, 43], equivalence checking [44], hardware mapping [41], and reasoning about quantum programs [23]. This relies on identifying a set of rewrite rules for a family of quantum circuits that transform a circuit into an equivalent one. Such a theory should be *sound* so that every rewrite preserves the underlying linear map. It should also be *complete* so that any equivalence between circuits can be derived. Additionally, we want it to be *minimal*, with no redundant rules.

Completeness results have been established for a wide range of circuit families, including reversible linear circuits [29, 38], reversible circuits over multiple-control Toffoli gates [24, 25], Clifford [9, 30, 34, 40], CNOT-dihedral [1, 14], two-qubit Clifford+T [11], three-qubit Clifford+CS [12], Toffoli–Hadamard [3], and real Clifford+CH [18] circuits, as well as universal quantum circuits [20, 21]. Completeness has also been obtained for the group of two-level matrices underlying several of these circuit families [10, 17, 23, 26, 31]. Minimal theories have been given for six of these circuit fragments [15] and for all quantum circuits [19].

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Completeness and minimality proofs typically require long case distinctions over gates, relations, and normal forms, whose supporting derivations sometimes span hundreds of pages [18, 23, 30]. In particular, relation reductions require intricate circuit manipulations and repeated rewrites, making them prone to human mistakes. For example, circular reasoning can lead us to remove a rule whose derivation still relies on that rule, hidden in a chain of other derivations. Errors can then propagate through dependent derivations, invalidating the entire reduction and making the error source hard to trace by hand.

The length of these proofs and their error-prone nature make it desirable to formally verify their correctness using a proof assistant like Lean or Agda. This would not have been feasible in the past due to the sheer number of lemmas and lengthy rewrites in these papers. Recent LLMs can write correct code for proof assistants, which brings this closer to reality. For example, Anthropic reports that LLMs wrote a 13-million-line Lean proof of Fermat's Last Theorem within 11 days [4]. However, care must still be taken. It is important to manually verify the *alignment* between the machine-verified statement and the intended natural-language result. For instance, [5] showed that a claimed verification of OpenAI's natural-language proof about the Navier–Stokes equations contained mistranslations, so the formal verification was verifying a weaker result by a different proof. A way to solve this is to use a hybrid approach, where humans write and carefully review the definitions and theorem statements [27], coding agents write the proofs, and Agda checks them.

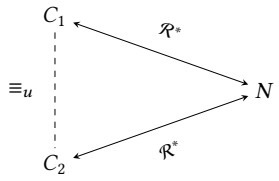
*Our contributions.* In this work we provide QCert, a human-constructed Agda [16, 37] framework for representing and verifying quantum circuit completeness and minimality results and use it to verify the correctness of sixteen completeness results. This builds on our formalization of qudit Clifford completeness [9], where we built the framework [6] and carried out the main verification steps by hand. Since all sixteen completeness results use the same framework of quantum circuits, gates, and matrices over a finitely generated ring, this makes it possible to efficiently check the alignment of the formalized results with the natural-language statements from the original papers. Of these sixteen results, we fully verify fourteen directly, and find fixes to verify the remaining two. For one result, we find a gap in its proof, which we fix, so the original completeness claim still holds. For another result, we find that the theorem is missing one relation, and we show that this relation is independent of the others, so completeness fails without it.

*Orientation.* In Section 2, we review the two main strategies for proving completeness of a rule set and how minimality is shown. In Section 3, we describe how QCert adds matrix semantics to our earlier Agda library of circuits and presentations [6], so that every completeness theorem is stated consistently as a group presentation. In Section 4, we present our formalization workflow and the techniques that keep typechecking fast. Then in Section 5, we demonstrate the usefulness of QCert by machine-checking the completeness and minimality results of sixteen papers. In Section 6, we briefly describe the issues these formalizations found in the source papers. In Section 8, we discuss the current limitations of QCert and our plans to extend it.

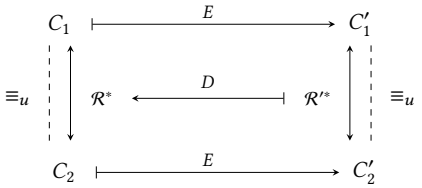
## 2 Background

A circuit is a word over an alphabet $\mathcal{G}$ of gates. A rewrite rule is then a relation between words over this alphabet, and a rule set $\mathcal{R}$ generates a congruence on circuits. To a circuit $C$ we associate an *interpretation* $[\![C]\!]$, which is its underlying matrix. The completeness results we formalize have matrix entries in the finitely generated rings $\mathbb{Z}\left[1/\sqrt{2}, i\right]$, $\mathbb{Z}\left[1/2, i\right]$, $\mathbb{Z}\left[1/\sqrt{2}\right]$, and $\mathbb{Z}\left[1/2\right]$, which make them exactly representable in Agda. We leave the formalization of universal circuits taking values in $\mathbb{C}$ for future work.

There are two main strategies for proving completeness of a rule set $\mathcal{R}$ over a gate set $\mathcal{G}$, both of which we support. Consider any two circuits $C_1$ and $C_2$ over $\mathcal{G}$ that have the same linear map, $[\![C_1]\!] = [\![C_2]\!]$. We need to show that $C_1$ can be rewritten into $C_2$ using $\mathcal{R}$. Figure 1a outlines the first strategy. It specifies a unique normal form for each linear map [1, 9, 14, 24, 25, 29, 30, 34, 38, 40], often by giving an exact synthesis algorithm [10, 17, 23, 26, 31]. Completeness follows by proving that every circuit can be transformed into its normal form using the rewrite rules. By uniqueness, $C_1$ and $C_2$ have the same normal form $N$. Reversing the path from $C_2$ to $N$ and concatenating it with the path from $C_1$ to $N$ then gives the desired path of rewrites.




(a) The rewrites from $C_1$ and $C_2$ to their common normal form $N$ using $\mathcal{R}$ give rewrites from $C_1$ to $C_2$ using $\mathcal{R}$.

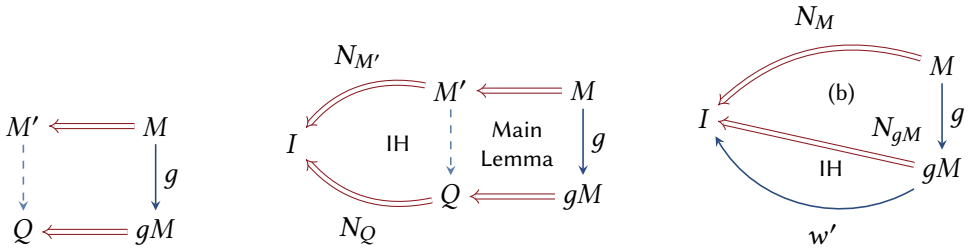



(b) Completeness of $\mathcal{R}'$ gives rewrites between $C_1' = E(C_1)$ and $C_2' = E(C_2)$. Translating this proof back gives rewrites between $C_1$ and $C_2$ using $\mathcal{R}$.

Fig. 1. Two strategies for proving completeness of $\mathcal{R}$ over a gate set $\mathcal{G}$. $\overset{\mathcal{R}^*}{\longleftrightarrow}$ means applying a finite sequence of rewrites using $\mathcal{R}$ in either direction, where $*$ denotes zero or more steps.

For the groups of two-level matrices, the normal forms come from exact synthesis, and the proof that every circuit can be transformed into its normal form has the same structure for each algorithm. Each matrix, or *state*, has a *level*, built from quantities such as the least denominator exponent of a column, and the algorithm reduces each state $M$ to the identity in steps that lower the level. These steps form the normal word $N_M$. The *Main Lemma* says that for every state $M$ and generator $g$, a diagram of relations closes the square formed by $g$ and the algorithm's next steps from $M$ and $gM$, passing only through states of lower level (Figure 2a). By well-founded induction on the level, the relations then rewrite $N_{gM}\, g$ into $N_M$ (Figure 2b), and by induction on length, every word that reduces $M$ to the identity rewrites into $N_M$ (Figure 2c). Two words with the same matrix therefore rewrite into each other. We formalize these inductions once and prove the Main Lemma for each presentation.

Figure 1b illustrates the second strategy, which uses an already complete rule set $\mathcal{R}'$ over a different gate set $\mathcal{G}'$ [3, 11–13, 15, 18, 20–22, 39]. Here, $E$ and $D$ translate circuits from




(a) The Main Lemma. (b) Induction on the level. (c) Induction on length.

Fig. 2. Completeness from the Main Lemma. Plain arrows are words, double arrows are steps or runs of the algorithm, and dashed arrows are words through states of lower level.

$\mathcal{G}$ to $\mathcal{G}'$ and back, replacing each gate with a circuit over the other gate set. Both $E$ and $D$ preserve circuit composition and the underlying linear map. To transfer completeness from $\mathcal{R}'$ to $\mathcal{R}$, we first show that every rule of $\mathcal{R}'$, translated by $D$, can be derived using $\mathcal{R}$. We can then replace each step of the proof between $C_1'$ and $C_2'$ by its translated derivation, giving $D(C_1') \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_2')$. It then remains to prove that the rewrite $C_j \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_j')$ is derivable, for which it suffices to prove the round-trip rewrites $g \overset{\mathcal{R}^*}{\longleftrightarrow} D(E(g))$ for each $g \in \mathcal{G}$. Sometimes the two strategies are combined, as in the scaled dyadic extension of [31].

Starting from a complete rule set, we can remove a redundant rule by deriving it from the other rules, thereby preserving completeness. To show that the resulting rule set is minimal, we need to give, for each remaining rule $R$, an interpretation of circuits that every other rule preserves but $R$ does not. Any derivation using the other rules preserves this interpretation, so $R$ cannot be derived from them [15, 19].

## 3 The QCert Framework

Our Agda framework QCert states every completeness theorem as a group presentation. Unlike the letters of an ordinary word, gates act on wires. The gates in $\mathcal{G}$ are indexed by the number of wires they act on. Wires are numbered from 0 at the bottom, and a gate always acts on the bottom wires of a circuit. To place it anywhere else, we apply a shift, which takes a circuit on $n$ wires and places it on wires $1, \ldots, n$ of a circuit with $n + 1$ wires. Each relation is stated on as many wires as the paper draws it on. To apply it inside a circuit with more wires, we need *structural rules*, which hold for every circuit, regardless of its gate set. These rules say that an equation can be shifted up by one wire, gates on disjoint wires commute, and a scalar can be placed on any wire. These define the equations of a PROP [32], and are often defined implicitly in completeness papers, but a formalization in Agda must define them explicitly.

We associate to circuits a matrix semantics $[\![-]\!]$ with entries in a finitely generated ring like $\mathbb{Z}\left[1/\sqrt{2}, i\right]$. Hence, $[\![-]\!]$ defines a group homomorphism to a unitary matrix group $G$. We build $G$ from the group structures of Agda's standard library, which is checked only once per ring. As matrices, a gate on the bottom wires and a shift are Kronecker

products with an identity, so the extension of $[\![-]\!]$ from gates to circuits is defined once for all gate sets. Since the structural rules are universal, their soundness is also proved only once. Then for each paper, it suffices to check the soundness of its own rules. Soundness and completeness say the relations present the image of $[\![-]\!]$, which is the subgroup of $G$ generated by the gates. In Agda, mathematical structures such as groups are defined as records. The record $\mathcal{R}$ IsSubPresentationOf $G$ contains the semantics $[\![-]\!]$, with proofs of soundness and completeness. The record $\mathcal{R}$ IsPresentationOf $G$ adds surjectivity, so the relations present all of $G$.

Each formalization ends with its main theorem stated as an IsSubPresentationOf or IsPresentationOf record. For example, Statement (1) is the theorem for the real stabilizer circuits of [34], with the relations R1–R16 of its Figure 1 lifted to $n$ wires. Statement (2) is the theorem for the two-level matrices of Hadamard-Π [23], with the relations of its Figure 6.

```
presentation : {n : ℕ} → (n VRel,_===_) IsSubPresentationOf U (2 ^ n)    (1)
presentation : {n : ℕ} → (_===_ {n}) IsPresentationOf (O n)              (2)
```

Agda checks every proof, so a reader only needs to check that each theorem statement matches its paper's result. Since every theorem is one of these two records, it suffices for a reader to check only four components: the relations transcribed from the paper, the target group, the matrix of each gate, and the line assigning $[\![-]\!]$ to the record. All sixteen statements are collected in a single root module, so the reader can check them in one place.

## 4 Formalization Workflow

Our workflow has three parts. First, before formalizing a paper, we check its rewrite rules, its intermediate results, and each line of its derivations with exact matrices in short scripts. This caught most typos before any Agda code was written. Second, we state each main theorem in QCert before a coding agent starts, so the agent only supplies the proof. Finally, for long derivations, the agent writes a script that turns each line of the paper into one step of the Agda proof, so a failing step points to a specific line of the paper.

The main computational cost came from Agda's conversion checker. If two terms are syntactically identical (i.e., written in the same way), it accepts them immediately. Otherwise, it evaluates both, which can take exponential time for a long matrix product. For such products, we keep circuit interpretations abstract and decide matrix equality by computing a single Boolean using row operations, which is orders of magnitude faster than comparing matrices entry by entry. We record these techniques in the instructions followed by the agents, so later proofs avoid the same slowdowns.

## 5 Formalized Completeness and Minimality Proofs

The sixteen formalizations in Table 1 contain about 420,000 lines of Agda code and use no postulates, and each theorem is as general as the result in its source paper. Moreover, all matrices are computed exactly. Their entries lie in algebraic number rings, whose elements are given by finitely many integers, so Agda checks automatically whether two matrices are equal. Over the real or complex numbers, no such automatic check exists, which limits the equality checking of QCert to discrete gate sets, as we discuss in Section 8.

| Source | Object | Result | Mod | Time |
|---|---|---|---|---|
| *Groups of m-level matrices* | | | | |
| Greylyn [2014] | $U_n(\mathbb{Z}[1/\sqrt{2}, i])$ | presentation, $n \leq 4$ | 31 | 42 |
| Bian and Selinger [2021] | $U_n(\mathbb{Z}[1/2, i])$ | presentation | 37 | 62 |
| Li et al. [2021] | $O_n(\mathbb{Z}[1/2])$, $L_n$ | presentations | 68 | 79 |
| Fang et al. [2026] | $O_n(\mathbb{Z}[1/\sqrt{2}])$ | presentation | 44 | 44 |
| Clément [2019] | $O_n(\mathbb{Z}[1/\sqrt{2}])$ | presentation | 40 | 64 |
| *Circuits* | | | | |
| Selinger [2015] | Clifford | sub-presentation | 31 | 34 |
| Bian, Li et al. [2026c] | Clifford, odd prime $p$ | presentation of a model | 215 | 546 |
| Makary et al. [2021] | real stabilizer | sub-presentation | 30 | 35 |
| Amy et al. [2018] | CNOT-dihedral | sub-presentation | 30 | 24 |
| Blake [2026b] | phase-affine, prime $d$ | sub-presentations | 56 | 61 |
| Bian and Selinger [2023a] | two-qubit Clifford+T | sub-presentation in $U_4$ | 62 | 118 |
| Bian and Selinger [2023b] | three-qubit Clifford+CS | sub-presentation in $U_8$ | 123 | 250 |
| Clément [2026] | real Clifford+CH | sub-presentation | 346 | 405 |
| Amy et al. [2024] | three-qubit Toffoli–Hadamard | presentation of $L_8$ | 201 | 102 |
| Li et al. [2025] | qutrit Clifford | presentation | 89 | 282 |
| *Minimality* | | | | |
| Blake [2026c] | six fragments | minimal, complete | 73 | 183 |
| Total | | | 1,476 | 2,332 |

Table 1. Completeness and minimality results formalized in QCert. A presentation gives generators and relations for a whole group, and a sub-presentation for the subgroup generated by the gate set named in the Object column of the Circuits block. $L_n$ is the group of orthogonal matrices $N/\sqrt{2}^j$, where $N$ is an integer matrix and $j \geq 0$. Mod denotes the number of Agda modules, the units that Agda checks one at a time. Time records their typechecking time in seconds from scratch, measured with Agda 2.8 on an Intel Core i5-14400. Mod and Time consider only each row's own modules, not the shared library or the formalizations it builds on.

Most theorems in Table 1 are proved from scratch in QCert against a matrix semantics, with three exceptions. The two rows of [11, 12] build on the Agda proofs that accompany those papers. These proofs relate the circuit relations to two-level relations by the Reidemeister–Schreier method [33]. We connect them to our formalizations of the two-level matrix presentations in [10, 26], the first two rows of the table, to obtain sub-presentations of $U_4\left(\mathbb{Z}\left[1/\sqrt{2}, i\right]\right)$ and $U_8\left(\mathbb{Z}\left[1/2, i\right]\right)$. The qudit Clifford theorem of [9] is the only one whose semantics is not a matrix group. Its group presentation lifts a presentation of the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_p)$ with Pauli corrections and global phases, and its semantics is built in the same way.

## 6 Corrections to the Source Papers

The formalizations in Table 1 identify two kinds of issues in the source papers. Most are typos, and some of them appear in statements that the proofs rely on. These are easy to fix.

Two others are harder to repair. One is a gap in a proof, and the other is a relation missing from a theorem. We describe both below.

The gap is in the proof of Theorem 4.14 of Hadamard-Π [23], which states that the relations in its Figure 6 present the group $O_n\left(\mathbb{Z}\left[1/\sqrt{2}\right]\right)$. One case of the proof relies on Lemmas A.16 and A.17, which have counterexamples in dimension 6, and the case analysis built on them misses a matrix in dimension 9. The authors have confirmed this gap (personal communication). Their main result is however still correct: Clément's unpublished notes [17] present the same group with different relations, and his case analysis covers the missing case, using which the formalization of Theorem 4.14 is completed. We also prove that the two sets of relations are equivalent in every dimension and formalize Clément's proof [7].

The relation is missing from Theorem 5.2 of [3]. It states that the relations in its Figures 4 and 7, over the Toffoli and $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ gates, present the group $O_8\left(\mathbb{Z}\left[1/2\right]\right)$. This is the first of two steps towards presenting the scaled dyadic orthogonal group $L_8$ by three-qubit Toffoli–Hadamard circuits. The relation $r_{\mathrm{K}}$ in Section 5.5, $\mathrm{K}_{1,2} = \mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0\,\mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0$, is in neither figure. Nor can it be derived from them. Hence, Theorem 5.2 fails as stated, and we have informed the authors. Our formalization [8] shows that $r_{\mathrm{K}}$ is independent by giving an interpretation of circuits that every other relation preserves but $r_{\mathrm{K}}$ does not, and proves the presentations of $O_8\left(\mathbb{Z}\left[1/2\right]\right)$ and $L_8$ with $r_{\mathrm{K}}$ restored.

## 7 A New Result: Toffoli–Hadamard Completeness for Any Number of Qubits

In 2021, Li et al. [31] found a complete set of rewrite rules for the group $O_n\left(\mathbb{Z}\left[1/2\right]\right)$, whose generators are one-, two-, and four-level matrices. These generators are matrices rather than gates, so the rules did not yet apply to circuits. In 2024, Amy et al. [3] observed that on three qubits, the gates X, CX, CCX, and $\mathrm{H} \otimes \mathrm{H}$ generate a finite group, the automorphism group of the $E_8$ lattice. They first turned its well-known Coxeter presentation into rewrite rules for these circuits. They then used the rules of Li et al. [31] for $n = 8$ to add the gates CCZ and H. This gave a complete set of rewrite rules for three-qubit Toffoli–Hadamard circuits, once a missing relation is restored (Section 6). Their method stops at three qubits. On four or more qubits, the same gates generate an infinite group, so there is no finite group to start from.

Recently, Clément [18] found a complete set of rewrite rules on any number of qubits for a closely related family, the real Clifford+CH circuits over the gates H, Z, CZ, and the controlled Hadamard gate CH. His rules consist of eighteen equations on at most three qubits and one equation schema, which gives an equation on $n$ qubits for each $n \geq 5$. His proof translates circuits back and forth to a presentation by one- and two-level matrices, as in Figure 1b. Following this approach with the presentations of Li et al. [31], we use QCert to generalize the result of Amy et al. [3] to any number of qubits, and Agda checks every step of the proof. In Section 7.1, we recall the gates, the matrix groups, and the presentations of Li et al. [31] that our proof builds on.

### 7.1 Preliminaries

The Toffoli–Hadamard gate set is {X, CX, CCX, H}, where CCX is the Toffoli gate. As in [3], we also consider the Toffoli–K gate set {X, CX, CCX, K}, where $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$. Recall that

$$\mathrm{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathrm{K} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

We number the qubits from 0, with qubit 0 the most significant, so the basis state $|x_0 x_1 x_2\rangle$ of three qubits is the basis vector $e_{4x_0+2x_1+x_2}$. We write $\mathrm{X}_j$ and $\mathrm{H}_j$ for the X and H gates on qubit $j$, and $\mathrm{K}_{j,k}$ for $\mathrm{H}_j\mathrm{H}_k$.

The interpretations of these circuits are orthogonal matrices. We write $O_n(\mathbb{Z}[1/2])$ for the group of $n \times n$ orthogonal matrices with entries in the ring of dyadic rationals $\mathbb{Z}[1/2]$. We write $L_n$ for the group of $n \times n$ orthogonal matrices of the form $N/\sqrt{2}^j$, where $N$ is an integer matrix and $j \geq 0$. The first group is a subgroup of the second, of index 2 when $n$ is even [31]. On $n$ qubits, the interpretations of Toffoli–K circuits lie in $O_{2^n}(\mathbb{Z}[1/2])$, and those of Toffoli–Hadamard circuits lie in $L_{2^n}$. With ancillas, every matrix in these groups is the interpretation of some circuit [2]. Without ancillas, as in this section, the circuits may generate only a subgroup. Hence, we state completeness on $n$ qubits as a sub-presentation (see Table 1).

Li et al. [31] gave presentations of both groups in every dimension $n$. Their generators are *multi-level* matrices rather than gates. For an $m \times m$ matrix $M$ and indices $a_1 < \cdots < a_m$ in $\{0, \ldots, n-1\}$, the $m$-level matrix $M_{[a_1,\ldots,a_m]}$ is the $n \times n$ matrix that acts as $M$ on the basis vectors $e_{a_1}, \ldots, e_{a_m}$ and as the identity on the others. The generators of $O_n(\mathbb{Z}[1/2])$ are the one-, two-, and four-level matrices

$$(-1)_{[a]}, \qquad \mathrm{X}_{[a,b]}, \qquad \mathrm{K}_{[a,b,c,d]}, \qquad 0 \leq a < b < c < d < n,$$

and a finite list of relation schemas, stated independently of $n$, presents this group [31, Theorem 4.10]. For even $n$, adding the generator $\mathrm{I} \otimes \mathrm{H} = \mathrm{diag}(\mathrm{H}, \ldots, \mathrm{H})$ and four relations gives a presentation of $L_n$ [31, Theorem 5.7]. A few generators are gates. On three qubits, for example, the Toffoli gate controlled by qubits 0 and 1 is $\mathrm{X}_{[6,7]}$, and $\mathrm{I} \otimes \mathrm{H}$ is $\mathrm{H}_2$. Most generators, however, are not gates of either gate set. For instance, $\mathrm{K}_{[4,5,6,7]}$ applies $\mathrm{K}_{1,2}$ only when qubit 0 is in the state $|1\rangle$. So these presentations are not yet equational theories for circuits.

Amy et al. [3] turned these presentations into circuit presentations on three qubits. Their method relies on the structure of three-qubit circuits. On three qubits, Toffoli–K circuits generate a finite group, the automorphism group $W(E_8)$ of the $E_8$ lattice, which has a well-known Coxeter presentation. Starting from it, they used Tietze transformations to rewrite the presentation of Li et al. [31] at $n = 8$ into one over gates, giving a presentation of $O_8(\mathbb{Z}[1/2])$ (their Theorem 5.2, with $r_{\mathrm{K}}$ restored). They then added $\mathrm{H}_2$ to obtain a presentation of $L_8$ (their Theorem 6.3), following [31, Section 5]. This method does not extend to more qubits. On four or more qubits, Toffoli–K circuits generate an infinite group, so there is no finite group such as $W(E_8)$ to start from. In the rest of this section, we

instead translate the presentations of Li et al. [31] into circuits on any number of qubits, using the second strategy of Section 2 (Figure 1b).

## 8 Discussion and Future Work

We formalize the results of sixteen papers in QCert, using consistent circuit syntax, a unified definition of group presentations, and shared libraries for rings and their matrix groups. Since all formalizations use these definitions, presentations of the same group from different papers can be compared and proved equivalent. The same libraries, together with the induction that turns an exact synthesis algorithm into a completeness proof, can be reused for new algorithms. Only the case analysis over matrices and generators is specific to each algorithm. As the most error-prone part of the proof, it is where machine checking in QCert matters most.

At the moment, not every completeness result fits into QCert, as its scope is limited in two ways. First, it supports only discrete gate sets, so it does not cover completeness results for circuits with continuous parameters, such as rotations by arbitrary angles [19–21]. Second, control is not a circuit constructor in QCert, so it does not cover results that treat control as a primitive operation on circuits [13, 22, 39].

We plan to extend the infrastructure of QCert to more families of quantum circuits, including the two above. Completeness is still open for many discrete gate sets in higher-dimensional quantum systems, and their case analyses are often too large to check by hand. In particular, we plan to study the completeness of Clifford circuits in all dimensions with the help of QCert. Beyond completeness, the exact synthesis libraries can be reused for T-count optimization of multi-qubit Clifford+T circuits. Finally, we are translating the formalizations from Agda into Lean, to make them available to its larger community. This will also allow us to take advantage of Lean's rich mathematical library Mathlib [42].

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